Numerical Methods

* The function/expression of the form
$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n$$
where n is a positive integer and $a_0, a_1, a_2, \ldots, a_n$ are constants with $a_0 \neq 0$ is known as a polynomial in x of degree 'n'.

- The values of x which satisfy $f(x) = 0$ (i.e., values of x making $f(x)$ zero) are known as zeroes or roots of the polynomial $f(x)$. Geometrically, a root of $f(x)=0$ is that value of x where the graph of $y=f(x)$ crosses the x-axis.

- The equation of the form $f(x)=0$ is called Algebraic if $f(x)$ is purely a polynomial in x.
  If $f(x)$ contains some other functions such as logarithmic, exponential, trigonometric functions etc., then $f(x)=0$ is called a transcendental equation.

Eg: (i) $x^5 - 2x^4 + 3x^2 - x + 5 = 0$ is an algebraic equation.
(ii) $2x^2 + \log(x-1) + e^x + \sin x = 0$ is a transcendental equation.

- Simple root : A number ξ is a simple root of $f(x)=0$ if
$$f(\xi) = 0, \; f'(\xi) \neq 0.$$

Multiple root : A number ξ is a multiple root of multiplicity 'm' if
$$f(\xi) = f'(\xi) = \cdots = f^{(m-1)}(\xi) \quad \ni f^{(m)}(\xi) \neq 0.$$
i.e., $f(x) = (x-\xi)^m g(x) \; ; g(\xi) \neq 0.$

Eg: In the polynomial $(x-2)^3(x+4)$, the root $x=2$ has a multiplicity of 3, and $x=-4$ has multiplicity 1.

- The process of finding the roots of an equation is called the solution of that equation.

* Basic properties of equations :

(i) If $f(x)$ is exactly divisible by $x-a$, then 'a' is a root of $f(x)=0$.

(ii) Every polynomial equation of the nth degree has only 'n' roots (real or imaginary).

(iii) In an equation with real coefficients, imaginary roots occur in conjugate pairs i.e., if $\alpha+i\beta$ is a root of the equation $f(x)=0$, then $\alpha-i\beta$ must also be its root.

(iv) Every polynomial equation of odd degree has atleast one real root.

* Intermediate Value property :

If $f(x)$ is continuous in the interval $[a,b]$ and $f(a)$, $f(b)$ have opposite signs, then the equation $f(x)=0$ has at least one root between $x=a$ and $x=b$, or an odd number of real roots in the interval (a,b).

* Iterative method :

Numerical methods of finding approximate roots of the given equation is a repetitive type of process known as iteration process. In each step the result of the previous step is used and the process is carried out till we get the result to the desired degree of accuracy. All the numerical methods are only approximate techniques for the solution of any problem. Further, this method gives only one root at a time.

Ex: Consider $a_0x^2+a_1x+a_2=0$.

The successive approximations are given by

$$x_{K+1} = -\frac{(a_0x_K^2+a_2)}{a_1} \quad ; K=0,1,2,\ldots$$

* Initial approximations :

Initial approximations to the root are often known from the physical considerations of the problem. Otherwise graphical methods are generally used to obtain initial approximations to the root. Since the value of x, at which the graph of $y=f(x)$ intersects the x-axis gives the root of $f(x)=0$, any value in the neighborhood of this point may be taken as an initial approximation to the root. If the equation $f(x)=0$ can be conveniently written in the form $f_1(x)=f_2(x)$, then the point of intersection of the graph of $y=f_1(x)$ and $y=f_2(x)$ gives the root of $f(x)=0$ and therefore any value in the neighborhood of this point can be taken as an initial approximation to the root.

Another commonly used method to obtain the initial approximation to the root is based upon the intermediate value theorem.

Eg: Find an initial guess to find a root of equation

(i) $2x - \log_{10} x = 7$

Soln. $f(x) = 2x - \log_{10} x - 7$.

x	1	2	3	4
$f(x)$	-5	-3.301	-1.477	0.397

$f(3).f(4) < 0$. Hence any value in $(3,4)$ can be taken as initial guess.

(ii) $2x - 3\sin x - 5 = 0$

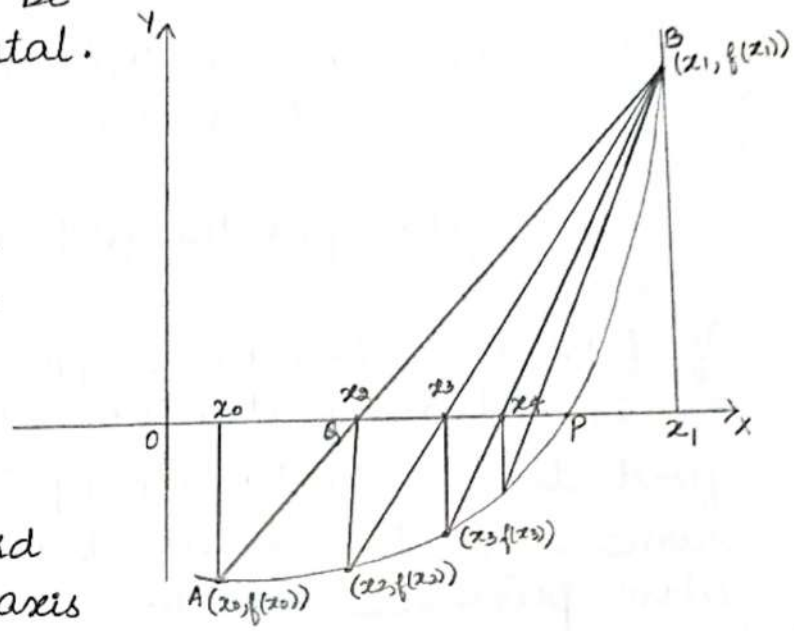
Soln $f(x) = 2x - 3\sin x - 5$

x	0	1	2	3
$f(x)$	-5	-5.5224	-3.7278	0.5766

$f(2).f(3) < 0$. Hence any value in $(2,3)$ can be taken as initial guess.

* Method of False position (or Regula - Falsi Method)

This is a method for finding the real root of an equation $f(x)=0$, where $f(x)$ may be algebraic or transcendental.



Let $f(x)=0$ be the given equation. Let $y=f(x)$ be represented by the curve AB. The real root of the equation $f(x)=0$ is OP. The false position of the curve AB is taken as chord AB. The chord AB cuts x-axis at Q. The approximate root of $f(x)=0$ is OQ. By this method we find OQ. Here we choose two points x_0 and x_1 such that $f(x_0)$ and $f(x_1)$ are of opposite sign. Thus the graph of $y=f(x)$ will cross x-axis between these two points showing that a root lies between x_0 and x_1 and consequently $f(x_0)f(x_1)<0$.

Now consider the equation of the chord joining the two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$.

$$y - f(x_0) = \frac{f(x_1)-f(x_0)}{x_1 - x_0}(x - x_0) \quad \longrightarrow (1)$$

(by coordinate geometry)

The Regula-Falsi method is based on replacing the part of the curve between the points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ by the chord joining these two points and then taking the point of intersection of the chord with x-axis as an approximate to the root i.e., put $y=0$ in (1) so that we obtain

$$-f(x_0) = \frac{f(x_1)-f(x_0)}{x_1 - x_0}(x_2 - x_0)$$

$$\Rightarrow \quad x_2 - x_0 = \frac{-f(x_0)(x_1 - x_0)}{f(x_1)-f(x_0)}$$

$$\Rightarrow \quad x_2 = x_0 - \frac{f(x_0)(x_1 - x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{x_0 f(x_1) - x_0 f(x_0) - x_1 f(x_0) + x_0 f(x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} \longrightarrow (2)$$

This formula gives the first approximation to the root.

If $f(x_2)$ and $f(x_0)$ are of opposite signs, then replace x_1 by x_2 and draw a straight line joining $f(x_2)$ and $f(x_0)$ to find the new intersection point. If $f(x_2)$ and $f(x_0)$ are of same sign then x_0 is replaced by x_2 and repeat the above procedure till the root is found to the desired accuracy.

Repeating the above process n times, we get the general formula given by

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

* <u>Note</u>:

1. Regula - Falsi method has linear rate of convergence.

2. Geometrically this method is equivalent to replacing the curve $y = f(x)$ by a chord that passes through the points $A[a, f(a)]$ and $B[b, f(b)]$ and intersection of chord with x-axis is obtained as solution of equation.

3. Taking $x_0 = a$ and $x_1 = b$ in (2), we obtain the formula

$$x = \frac{a f(b) - b f(a)}{f(b) - f(a)}$$

This formula is <u>used</u> while working the problems.

1. Solve for a positive root of $x^3-4x+1=0$ by Regula - Falsi method upto four decimal places.

Soln $f(x) = x^3-4x+1$

$f(0) = 1 > 0$; $f(1) = -2 < 0$

$\therefore$ Root lies in $(0,1)$

Iteration	a	b	f(a)	f(b)	$x = \frac{af(b)-bf(a)}{f(b)-f(a)}$	$f(x) = x^3-4x+1$
	A	B	C	D	$X = \frac{AD-BC}{D-C}$	$Y = X^3-4X+1$
1	0	1	1	-2	0.3333	-0.2963
2	0	0.3333	1	-0.2963	0.2571	-0.0115
3	0	0.2571	1	-0.0115	0.2542	-0.0003
4	0	0.2542	1	-0.0003	0.2541	-0.0001
5	0	0.2541	1	-0.0001	0.2541	

$\therefore$ The required root correct to 4 decimal places is $x = 0.2541$.

2. Using Regula falsi method, compute the real root of equation $xe^x = 2$ correct to 4 decimal places.

Soln $f(x) = xe^x - 2$

$f(0) = -2 < 0$; $f(1) = 0.7183 > 0$

$\therefore$ Root lies in $(0,1)$

Iteration	a	b	f(a)	f(b)	$x = \frac{af(b)-bf(a)}{f(b)-f(a)}$	$f(x) = xe^x - 2$
1	0	1	-2	0.7183	0.7358	-0.4644
2	0.7358	1	-0.4644	0.7183	0.8395	-0.0562
3	0.8395	1	-0.0562	0.7183	0.8511	-0.0063
4	0.8511	1	-0.0063	0.7183	0.8524	-0.0009
5	0.8524	1	-0.0009	0.7183	0.8526	-0.0001
6	0.8526	1	-0.0001	0.7183	0.8526	

$\therefore$ Required root correct to 4 decimal places is $x = 0.8526$.

3. Using Regula falsi method compute the real root of equation $\cos x = 3x - 1$ correct to 4 decimal places.

Soln $f(x) = \cos x - 3x + 1$

$f(0) = 2 > 0$; $f(1) = -1.4597 < 0$

$\therefore$ Root lies in $(0, 1)$

a	b	$f(a)$	$f(b)$	$x = \frac{af(b) - bf(a)}{f(b) - f(a)}$	$f(x) = \cos x - 3x + 1$
0	1	2	-1.4597	0.5781	0.1033
0.5781	1	0.1033	-1.4597	0.6060	0.004
0.6060	1	0.0040	-1.4597	0.6071	0.0001
0.6071	1	0.0001	-1.4597	0.6071	

$\therefore$ The required root correct to 4 decimal places is $x = 0.6071$

* Newton - Raphson method

Given an approximate value of a root of an equation, a better and closer approximation to the root can be found by using an iterative process known as Newton - Raphson method. Assuming that x_0 is an approximate value of a real root of the equation $f(x)=0$ ($f(x)$ may be Algebraic or Transcendental), let x_1 be the exact root and $x_1 = x_0 + h$ where h is small correction.

We have $f(x_1) = 0$ i.e., $f(x_0+h) = 0$.
Using Taylor's expansion, we write

$$f(x_0+h) = f(x_0) + h f'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots = 0$$

Since h is a small quantity, $h^2, h^3, \dots$ being still smaller can be neglected. Thus we have,

$$f(x_0) + h f'(x_0) = 0 \Rightarrow h = -\frac{f(x_0)}{f'(x_0)}$$

Using this in $x_1 = x_0 + h$, we obtain

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$ (provided $f'(x_0) \neq 0$) which is the first approximation.

Similarly starting with x_1, we get a better approximation x_2,

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$ and so on.

In general, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, $n = 0, 1, 2, \dots$

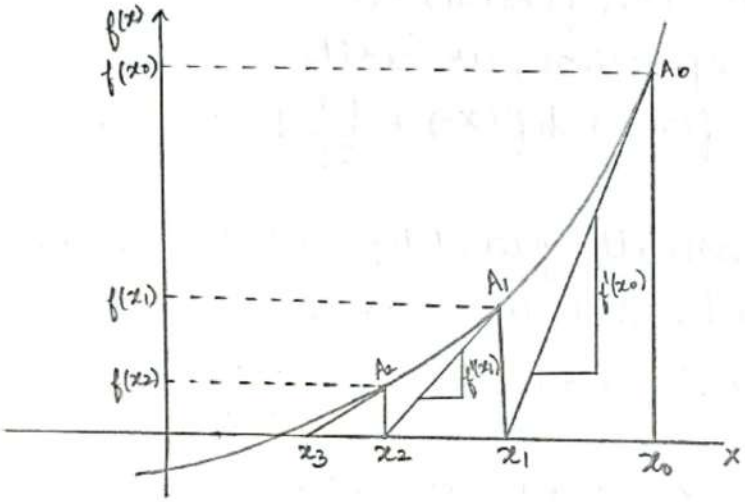
This is called Newton - Raphson iterative formula.

* Note :

1. This method has quadratic convergence. i.e., the subsequent error at each step is proportional to the square of the previous error.

2. This method will fail if $f'(x)=0$ in the neighborhood of the root.
3. If we choose x_0 close to the root then we get the root of the equation more rapidly.
4. If $f'(x)$ is very large, i.e., when the graph of the function when it crosses the x-axis nearly vertical then this method is very useful.
5. This method is also referred as the method of tangents.

* Geometrical Interpretation



Let x_0 be a point near the root α of the equation $f(x)=0$. Then, the equation of the tangent at $A_0(x_0, f(x_0))$ is

$$y - f(x_0) = f'(x_0)(x - x_0)$$

(Put $y=0$. Then $x = x_1$) The tangent cuts x-axis at

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad \rightarrow \text{ first approximation to root } \alpha.$$

If A_1 is a point corresponding to x_1 on the curve, then the tangent at A_1 will cut the x-axis at x_2, which is nearer to α is a second approximation to the root. Hence the method consists of replacing the part of the curve between the point A_0 and the x-axis by means of tangent to the curve at A_0.

1. Find the real root of equation $xe^x - 2 = 0$ correct to 4 decimal places using Newton-Raphson method.

Soln

$f(x) = xe^x - 2$

$f'(x) = xe^x + e^x = e^x(x+1)$

$f(0) = -2 < 0$; $f(1) = 0.7183 > 0$.

$\therefore$ Root lies in $(0, 1)$

Let $x_0 = 1$

n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
	$A = xe^x - 2$	$B = xe^x + e^x$	$x = x - \frac{A}{B}$
0	0.7183	5.4366	0.8679
1	0.0672	4.4490	0.8528
2	0.0008	4.3469	0.8526
3	0		

Since $f(0.8526) = 0$, the required root is $x = 0.8526$.

2. Use Newton-Raphson method to find $\sqrt[3]{37}$ correct to 4 decimal places.

Soln

Let $x = \sqrt[3]{37} \Rightarrow x^3 = 37 \Rightarrow x^3 - 37 = 0$

$f(x) = x^3 - 37$; $f'(x) = 3x^2$

$f(3) = -10 < 0$; $f(4) = 27 > 0$

$\therefore$ Real root lies in $(3, 4)$.

Let $x_0 = 3$

n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
	$A = x^3 - 37$	$B = 3x^2$	$x = x - A/B$
0	-10	27	3.3704
1	1.2854	34.0782	3.3327
2	0.0143	33.3197	3.3322
3	0		

Since $f(3.3322) = 0$, the required root is $x = 3.3322$

* Some deductions from the Newton-Raphson formula.

1. Reciprocal of a number :

Show that the iterative formula to find $\frac{1}{N}$ is $x_{n+1} = x_n(2 - Nx_n)$

Proof : Let $x = \frac{1}{N}$ or $\frac{1}{x} - N = 0$

Taking $f(x) = \frac{1}{x} - N$, we have $f'(x) = \frac{-1}{x^2} = -x^{-2}$

Then, Newton Raphson formula gives,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{\left(\frac{1}{x_n} - N\right)}{-x_n^{-2}} = x_n + \left(\frac{1}{x_n} - N\right)x_n^2$$

$$= x_n + \left(\frac{1 - Nx_n}{x_n}\right)x_n^2 = x_n + x_n - Nx_n^2$$

$$x_{n+1} = 2x_n - Nx_n^2$$

$$x_{n+1} = x_n(2 - Nx_n)$$

2. Square root of a number :

Show that the iterative formula to find $\sqrt{N}$ is $x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)$

Proof : Let $f(x) = x^2 - N$; $f'(x) = 2x$

$x = \sqrt{N}$
$x^2 = N$
$x^2 - N = 0$

Then, Newton Raphson formula gives,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^2 - N}{2x_n} = \frac{2x_n^2 - x_n^2 + N}{2x_n}$$

$$= \frac{x_n^2 + N}{2x_n}$$

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)$$

3. Reciprocal of the square root :

Show that the iterative formula to find $\frac{1}{\sqrt{N}}$ is $x_{n+1} = \frac{1}{2}\left(x_n + \frac{1}{Nx_n}\right)$

<u>Proof</u>: Let $x = \frac{1}{\sqrt{N}}$ or $x^2 - \frac{1}{N} = 0$

Taking $f(x) = x^2 - \frac{1}{N}$, we have $f'(x) = 2x$

Then, Newton-Raphson formula gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

i.e.,

$$x_{n+1} = x_n - \frac{\left(x_n^2 - \frac{1}{N}\right)}{2x_n} = \frac{2x_n^2 - x_n^2 + \frac{1}{N}}{2x_n} = \frac{x_n^2 + \frac{1}{N}}{2x_n}$$

$$= \frac{1}{2}\left(x_n + \frac{1}{Nx_n}\right)$$

4. K^{th} root of a number

Show that the iterative formula to find $\sqrt[K]{N}$ is

$$\frac{1}{K}\left[(K-1)x_n + \frac{N}{x_n^{K-1}}\right]$$

<u>Proof</u>: Let $x = \sqrt[K]{N}$ or $x^K - N = 0$

Taking $f(x) = x^K - N$, we have $f'(x) = Kx^{K-1}$

Then, Newton-Raphson formula gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^K - N}{Kx_n^{K-1}} = \frac{Kx_n^K - x_n^K + N}{Kx_n^{K-1}}$$

$$= \frac{1}{K}\left[\frac{(K-1)x_n^K + N}{x_n^{K-1}}\right]$$

$$= \frac{1}{K}\left[(K-1)x_n + \frac{N}{x_n^{K-1}}\right]$$

5. Reciprocal of the Kth root

Show that the iterative formula to find $\frac{1}{\sqrt[K]{N}}$ is $\frac{x_n}{K}\left[K+1-Nx_n^K\right]$.

Proof: Let $x = \frac{1}{N^{1/K}}$, Then $f(x) = N - x^{-K}$

$$f'(x) = Kx^{-(K+1)}$$

Newton - Raphson formula gives,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{(N - x_n^{-K})}{Kx_n^{-(K+1)}}$$

$$= x_n - \frac{(Nx_n^{K+1} - x_n)}{K} = \frac{x_nK - Nx_n^{K+1} + x_n}{K}$$

$$x_{n+1} = \frac{x_n(K+1) - Nx_n^{K+1}}{K}$$

$$= \frac{x_n}{K}\left[K+1 - Nx_n^K\right]$$

Examples

1. Evaluate $\frac{1}{31}$ correct to four decimal places using Newton - Raphson method.

Soln Newton - Raphson iteration formula to find $\frac{1}{N}$ is $x_{n+1} = x_n(2 - Nx_n)$

$\therefore N = 31$.

Since an approximate value of $\frac{1}{31}$ is 0.03, we take $x_0 = 0.03$.

$$x_1 = x_0(2 - Nx_0)$$
$$= 0.03(2 - 31(0.03)) = 0.0321$$

$$x_2 = x_1(2 - Nx_1)$$
$$= 0.0321\left[2 - 31(0.0321)\right]$$
$$= 0.0323$$

$$x_3 = x_2(2 - Nx_2)$$
$$= 0.0323[2 - 31(0.0323)] = 0.0323$$

Since $x_2 = x_3$ upto four decimal places, we have $\frac{1}{31} = 0.0323$

2. Develop an algorithm using Newton-Raphson method to find $\frac{1}{\sqrt{N}}$ and hence compute to three/four decimal places $\frac{1}{\sqrt{15}}$.

Soln Refer the previous section for algorithm / recurrence relation / iterative formula.

We have $x_{n+1} = \frac{1}{2}\left(x_n + \frac{1}{Nx_n}\right)$, $N = 15$

Since approximate value of $\frac{1}{\sqrt{15}}$ is $\frac{1}{\sqrt{16}} = \frac{1}{4} = 0.25$, we take

$x_0 = 0.2$

$$x_1 = \frac{1}{2}\left(x_0 + \frac{1}{Nx_0}\right)$$
$$= \frac{1}{2}\left[0.2 + \frac{1}{15(0.2)}\right] = 0.2667$$

$$x_2 = \frac{1}{2}\left(x_1 + \frac{1}{Nx_1}\right)$$
$$= \frac{1}{2}\left[0.2667 + \frac{1}{15(0.2667)}\right] = 0.2583$$

$$x_3 = \frac{1}{2}\left(x_2 + \frac{1}{Nx_2}\right)$$
$$= \frac{1}{2}\left[0.2583 + \frac{1}{15(0.2583)}\right] = 0.2582$$

$$x_4 = \frac{1}{2}\left(x_3 + \frac{1}{Nx_3}\right)$$
$$= \frac{1}{2}\left[0.2582 + \frac{1}{15(0.2582)}\right] = 0.2582$$

Since $x_3 = x_4$ upto four decimal places, we have $\frac{1}{\sqrt{15}} = 0.2582$

3. Evaluate $(28)^{-1/4}$ using N-R method correct to 4 decimal places.

Soln

$(28)^{-1/4} = \frac{1}{(28)^{1/4}}$ is of the form $\frac{1}{(N)^{1/K}}$

$\therefore N = 28$, $K = 4$.

Note that $(16)^{-1/4} = \frac{1}{2}$ and $(81)^{-1/4} = \frac{1}{3}$

16 is nearer to 28. Hence we take $x_0 = \frac{1}{2}$ as initial approximation

We have $x_{n+1} = \frac{x_n}{K}\left[K+1 - N x_n^K\right]$ - iterative formula to find $\frac{1}{(N)^{1/K}}$.

$\therefore x_1 = \frac{x_0}{K}\left[K+1 - N x_0^K\right]$

$= \frac{1}{8}\left[4+1 - 28\left(\frac{1}{2}\right)^4\right] = 0.4063$

$x_2 = \frac{x_1}{K}\left[K+1 - N x_1^K\right] = 0.4304$

$x_3 = \frac{x_2}{K}\left[K+1 - N x_2^K\right] = 0.4346$

$x_4 = \frac{x_3}{K}\left[K+1 - N x_3^K\right] = 0.4347$

$x_5 = \frac{x_4}{K}\left[K+1 - N x_4^K\right] = 0.4347$

Since $x_4 = x_5$ upto four decimal places, $\underline{(28)^{-1/4} = 0.4347}$.

* Numerical Solution of Ordinary differential equations.

In the fields of Engineering and Science, we come across physical and natural phenomena which when represented by mathematical models happen to be differential equations. Sometimes analytical solutions maynot exist and in some cases it may be very difficult to solve by analytic methods. In such cases numerical methods play an important role.

A general solution of a differential equation of the nth order has n arbitrary constants. In order to compute numerical solution of such an equation we need n conditions. If all the n conditions are prescribed at the initial point (one point) only, then it is called initial value problem. If the n conditions are specified at two or more points, then it is called boundary value problem.

* Taylor's Series method

To find the numerical solution of the equation $\frac{dy}{dx} = f(x,y)$

with the initial condition $y(x_0) = y_0$. The solution $y(x)$ is approximated to power series in $(x - x_0)$ using Taylor's series. Then we can find the value of y for various values of x in the neighborhood of x_0. We have Taylor's series expansion of $y(x)$ about the point x_0 in the form:

$$y(x) = y(x_0) + \frac{(x-x_0)}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \cdots$$

Taylor series solution at the nth step is given by,

$$y_{n+1} = y_n + \frac{h}{1!} y'_n + \frac{h^2}{2!} y''_n + \frac{h^3}{3!} y'''_n + \cdots$$

where $h = x_{n+1} - x_n$; $y_n = y(x_n)$, $y'_n = y'(x_n)$, ...

1. Use Taylor's series method to find y at points $x_1 = 0.1$ and $x_2 = 0.2$, given that $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$.

Soln Here $x_0 = 0$, $y_0 = 1$

$$y' = x^2 + y^2 \quad ; \quad y'(0) = 0 + 1 = 1$$
$$y'' = 2x + 2yy' \quad ; \quad y''(0) = 2$$
$$y''' = 2 + 2\{yy'' + y'^2\} \quad ; \quad y'''(0) = 2 + 2(2+1) = 8$$
$$y^{IV} = 2\{yy''' + y'y'' + 2y'y''\} \quad ; \quad y^{IV}(0) = 2(8+6) = 28$$
$$= 2\{yy''' + 3y'y''\}$$

Taylor's series solution is given by,

$$y(x) = y(x_0) + \frac{(x-x_0)}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \cdots$$

$$= y(0) + x y'(0) + \frac{x^2}{2!} y''(0) + \frac{x^3}{3!} y'''(0) + \frac{x^4}{4!} y^{IV}(0)$$

$$= 1 + x(1) + \frac{x^2}{2}(2) + \frac{x^3}{6}(8) + \frac{x^4}{24}(28)$$

$$y(x) = 1 + x + x^2 + \frac{4}{3}x^3 + \frac{7}{6}x^4$$

$$\therefore y(0.1) = 1 + (0.1) + (0.1)^2 + \frac{4}{3}(0.1)^3 + \frac{7}{6}(0.1)^4 = 1.1115$$

$$y(0.2) = 1 + (0.2) + (0.2)^2 + 4/3(0.2)^3 + 7/6(0.2)^4 = 1.2525$$

2. Find y at $x = 0.1$ correct to four decimal places given $\frac{dy}{dx} = x - y^2$; $y(0) = 1$ applying Taylor's series method upto fourth degree term.

Soln $y' = x - y^2 \quad ; \quad y'(0) = -1 \qquad x_0 = 0, \; y_0 = 1$

$$y'' = 1 - 2yy' \quad ; \quad y''(0) = 3$$

$$y''' = -2[yy'' + (y')^2] \quad ; \quad y'''(0) = -2[3+1] = -8$$

$$y^{IV} = -2\left[yy''' + y''y' + 2y'y''\right] = -2\left[yy''' + 3y'y''\right]$$

$$y^{IV}(0) = -2\left[-8 + 3(-1)(3)\right] = 34$$

Taylor's series solution is given by,

$$y(x) = y(x_0) + (x - x_0)y'(x_0) + \frac{(x-x_0)^2}{2!}y''(x_0) + \cdots$$

$$= y(0) + xy'(0) + \frac{x^2}{2!}y''(0) + \frac{x^3}{3!}y'''(0) + \frac{x^4}{4!}y^{IV}(0)$$

$$= 1 + x(-1) + \frac{x^2}{2}(3) + \frac{x^3}{6}(-8) + \frac{x^4}{24}(34)$$

$$y(x) = 1 - x + \frac{3x^2}{2} - \frac{4x^3}{3} + \frac{17x^4}{12}$$

$$\therefore y(0.1) = 0.9138$$

3. Use Taylor's series method to find y at $x = 0.1$ and $x = 0.2$ given $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$

Soln

$$y' = x^2y - 1 \quad ; \quad y'(0) = -1$$

$$y'' = 2xy + x^2y' \quad ; \quad y''(0) = 0$$

$$y''' = 2y + 2xy' + 2xy' + x^2y'' \quad ; \quad y'''(0) = 2$$

$$y^{IV} = 2y' + 4y' + 4xy'' + 2xy'' + x^2y''' \quad ; \quad y^{IV}(0) = 2(-1) + 4(-1) = -6$$

Taylor's series solution is given by,

$$y(x) = y(x_0) + (x - x_0)y'(x_0) + \frac{(x-x_0)^2}{2!}y''(x_0) + \cdots$$

$$= y(0) + xy'(0) + \frac{x^2}{2!}y''(0) + \frac{x^3}{3!}y'''(0) + \frac{x^4}{4!}y^{IV}(0)$$

$$= 1 + x(-1) + \frac{x^2}{2!}(0) + \frac{x^3}{6}(2) + \frac{x^4}{24}(-6)$$

$$y(x) = 1 - x + \frac{x^3}{3} - \frac{x^4}{4}$$

$$\therefore y(0.1) = 1 - (0.1) + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} = 0.9003$$

$$\text{III}^{ly}\ y(0.2) = 0.8023$$

4. Employ Taylor's method to obtain approximate value of y at $x = 0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x$; $y(0) = 0$.

Soln

$$y' = 2y + 3e^x \quad ; \; y'(0) = 3$$
$$y'' = 2y' + 3e^x \quad ; \; y''(0) = 2(3) + 3 = 9$$
$$y''' = 2y'' + 3e^x \quad ; \; y'''(0) = 2(9) + 3 = 21$$
$$y^{iv} = 2y''' + 3e^x \quad ; \; y^{iv}(0) = 2(21) + 3 = 45$$

Substituting these values in Taylor's series, we have

$$y(x) = y(0) + xy'(0) + \frac{x^2}{2!}y''(0) + \frac{x^3}{3!}y'''(0) + \frac{x^4}{4!}y^{iv}(0)$$

$$= 0 + x(3) + \frac{x^2}{2}(9) + \frac{x^3}{6}(21) + \frac{x^4}{24}(45)$$

$$y(x) = 3x + \frac{9}{2}x^2 + \frac{7}{2}x^3 + \frac{45}{24}x^4$$

$$\therefore \; y(0.2) = \underline{\underline{0.811}}$$

* Runge - Kutta method of fourth order

This method which is most widely used, was first devised by C. Runge in 1894 and then extended by W. Kutta in 1901. In Runge - Kutta methods, the derivatives of higher order are not required and we require only the given function values at different points. In this method, we calculate the increments of the function by means of a definite set of formulae.

A set of formulae are given for solving a differential equation of the form $\frac{dy}{dx} = f(x,y)$ under the initial condition $y(x_0) = y_0$.

Let h denote length of the interval between equidistant values of x. The approximate solution of the equation at $x_{n+1} = x_n + h$ is given by

$$y_{n+1} = y(x_n + h) = y_n + \frac{1}{6}\left[K_1 + 2K_2 + 2K_3 + K_4\right]$$

where

$$K_1 = h\, f(x_n, y_n)$$

$$K_2 = h \cdot f\left(x_n + \frac{h}{2}, y_n + \frac{K_1}{2}\right)$$

$$K_3 = h\, f\left(x_n + \frac{h}{2}, y_n + \frac{K_2}{2}\right)$$

$$K_4 = h\, f(x_n + h, y_n + K_3)$$

(or)

$$y_{n+1} = y_n + K$$

where $K = \frac{1}{6}\left[K_1 + 2K_2 + 2K_3 + K_4\right]$

1. Using fourth order Runge-Kutta method, find $y(0.2)$ for the equation $\frac{dy}{dx} = 3x + \frac{y}{2}$ with $y(0) = 1$ taking $h = 0.2$.

Soln Here, $f(x,y) = 3x + \frac{y}{2}$; $x_0 = 0$, $y_0 = 1$, $h = 0.2$

We need to compute $y_1 = y(0.2) = y(x_0+h) = y_0 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$

$$K_1 = hf(x_0, y_0) = 0.2\, f(0,1) = 0.2\left[3(0) + \frac{1}{2}\right] = 0.1$$

$$K_2 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right) = 0.2\, f\left(\frac{0.2}{2}, 1 + \frac{0.1}{2}\right) = 0.2\, f(0.1, 1.05)$$

$$= 0.2\left[3(0.1) + \frac{1.05}{2}\right] = 0.165$$

$$K_3 = h\, f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right) = 0.2\, f\left(0 + \frac{0.2}{2}, 1 + \frac{0.165}{2}\right) = 0.2\, f(0.1, 1.0825)$$

$$= 0.2\left[3(0.1) + \frac{1.0825}{2}\right] = 0.1683$$

$$K_4 = h\, f(x_0 + h, y_0 + K_3) = 0.2\, f(0 + 0.2, 1 + 0.1683)$$

$$= 0.2\left[3(0.2) + \frac{1.1683}{2}\right] = 0.2368$$

$$\therefore \quad y(0.2) = y_0 + \frac{1}{6}\left[K_1 + 2K_2 + 2K_3 + K_4\right]$$

$$= 1 + \frac{1}{6}\left[0.1 + 2(0.165) + 2(0.1683) + 0.2368\right]$$

$$= 1 + 0.1672$$

$$= 1.1672$$

2. Solve $\frac{dy}{dx} = 1 - 2xy$; $y(0) = 0$; find the approximate solution for $x = 0.1\,(0.1)\,0.2$

Sol. Here, $f(x,y) = 1 - 2xy$; $x_0 = 0$, $y_0 = 0$, $h = 0.1$

We need to compute $y_1 = y(0.1)$ and $y_2 = y(0.2)$.

$$y_1 = y_0 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

where $K_1 = hf(x_0, y_0) = 0.1 f(0,0) = 0.1$

$$K_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right) = 0.1 f\left(0 + \frac{0.1}{2}, 0 + \frac{0.1}{2}\right) = 0.0995$$

$$K_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right) = 0.1 f\left(0 + \frac{0.1}{2}, \frac{0.0995}{2}\right) = 0.0995$$

$$K_4 = hf(x_0 + h, y_0 + K_3) = 0.1 f(0 + 0.1, 0.0995) = 0.098$$

$$y_1 = y(0.1) = y_0 + K = 0 + \frac{1}{6}\left[0.1 + 2(0.0995) + 2(0.0995) + 0.098\right]$$

$$= \underline{\underline{0.0993}}$$

$$y_2 = y_1 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

where $K_1 = hf(x_1, y_1) = 0.1 f(0.1, 0.0993) = 0.098$

$$K_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{K_1}{2}\right) = 0.1 f\left(0.1 + \frac{0.1}{2}, 0.0993 + \frac{0.098}{2}\right) = 0.0956$$

$$K_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{K_2}{2}\right) = 0.1 f\left(0.1 + \frac{0.1}{2}, 0.0993 + \frac{0.0956}{2}\right) = 0.0956$$

$$K_4 = hf(x_1 + h, y_1 + K_3) = 0.1 f(0.1 + 0.1, 0.0993 + 0.0956)$$

$$= 0.0922$$

$$\therefore y_2 = 0.0993 + \frac{1}{6}\left[0.098 + 2(0.0956) + 2(0.0956) + 0.0922\right]$$

$$= 0.1947$$

$$\therefore y(0.2) = \underline{\underline{0.1947}}$$

* Predictor - Corrector Methods:

The methods discussed earlier to solve a first order differential equation, require information about the solution at a single point say x_n for computing the solution at x_{n+1}. Hence these methods are called single-step methods.

In the Predictor - Corrector methods, a predictor formula is used to predict the value y_{n+1} of y at x_{n+1} and then a corrector formula is used to improve the value of y_{n+1}. The corrector formula is used repeatedly until two consecutive values of y_{n+1} are almost equal.

* Milne's Predictor - Corrector method

The Predictor - Corrector methods form a large class of general methods for numerical integration of ordinary differential equations.

Consider the differential equation

$$\frac{dy}{dx} = f(x,y) \quad \text{with initial condition } y(x_0) = y_0.$$

Define

$$y_1 = y(x_0+h)$$
$$y_2 = y(x_0+2h)$$
$$\vdots$$
$$y_n = y(x_0+nh)$$

$$y'(x) = f(x,y)$$
$$y_0'(x) = f(x_0,y_0)$$
$$y_1'(x) = f(x_0+h, y_1) = f(x_1,y_1)$$
$$y_2'(x) = f(x_0+2h, y_2) = f(x_2,y_2)$$
$$\vdots$$
$$y_n'(x) = f(x_0+nh, y_n) = f(x_n,y_n)$$

Predictor formula :

$$y_4^{(P)} = y_0 + \frac{4h}{3}\left(2y_1' - y_2' + 2y_3'\right)$$

$$y_5^{(P)} = y_1 + \frac{4h}{3}\left(2y_2' - y_3' + 2y_4'\right)$$

In general,

$$y_{n+1}^{(P)} = y_{n-3} + \frac{4h}{3}\left[2y_{n-2}' - y_{n-1}' + 2y_n'\right]$$

(or)

$$y_{n+1}^{(P)} = y_{n-3} + \frac{4h}{3}\left[2f_{n-2} - f_{n-1} + 2f_n\right]$$

Corrector formula :

$$y_4^{(C)} = y_2 + \frac{h}{3}\left(y_2' + 4y_3' + y_4'\right)$$

$$y_5^{(C)} = y_3 + \frac{h}{3}\left(y_3' + 4y_4' + y_5'\right)$$

In general,

$$y_{n+1}^{(C)} = y_{n-1} + \frac{h}{3}\left[y_{n-1}' + 4y_n' + y_{n+1}'\right]$$

(or)

$$y_{n+1}^{(C)} = y_{n-1} + \frac{h}{3}\left[f_{n-1} + 4f_n + f_{n+1}\right]$$

The need for a corrector formula arises because the predictor alone is numerically unstable ; it gives spurious solutions growing exponentially.

Milne's predictor uses four previous values of y, these values can be obtained by using Taylor's or Runge-Kutta method.

Note: Improved value of $y_4^{(P)}$ is calculated using $y_4^{(C)}$ and again the corrector is applied to find a still better value of $y_4^{(C)}$. This step is repeated till two consecutive values of $y_4^{(C)}$ are same.

Once y_4 and y_4' are obtained to the desired degree of accuracy, the next value of y is obtained from predictor-corrector equations.

1. Using Milne's predictor-corrector method, evaluate $y(0.4)$ given $\frac{dy}{dx} = \frac{1}{2}(1+x^2)y^2$ and $y(0)=1$, $y(0.1)=1.06$, $y(0.2)=1.12$, $y(0.3)=1.21$

Soln Here, $y' = f(x,y) = \frac{1}{2}(1+x^2)y^2$; $h=0.1$

x	y	$y' = f(x,y)$
$x_0 = 0$	$y_0 = 1$	$y_0' = f_0 = f(x_0,y_0) = 0.5$
$x_1 = 0.1$	$y_1 = 1.06$	$y_1' = f_1 = f(x_1,y_1) = 0.5674$
$x_2 = 0.2$	$y_2 = 1.12$	$y_2' = f_2 = f(x_2,y_2) = 0.6523$
$x_3 = 0.3$	$y_3 = 1.21$	$y_3' = f_3 = f(x_3,y_3) = 0.7979$
$x_4 = 0.4$	$y_4 = ?$	

By Milne's predictor method,

$$y_4^{(P)} = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3')$$

$$= 1 + \frac{4(0.1)}{3}\left[2(0.5674) - 0.6523 + 2(0.7979)\right]$$

$$= 1 + 0.2771 = 1.2771$$

$$y_4'^{(P)} = 0.9460 = f(x_4, y_4^{(P)}) = f(0.4, 1.2771)$$

By Milne's corrector method,

$$y_4^{(C)} = y_2 + \frac{h}{3}(y_2' + 4y_3' + y_4'^{(P)})$$

$$= 1.12 + \frac{0.1}{3}\left[0.6523 + 4(0.7979) + 0.9460\right] = 1.12 + 0.1597$$

$$= 1.2797$$

Again $y_4'^{(c)} = f(x_4, y_4^{(c)}) = f(0.4, 1.2797) = 0.9498$

Second approximation of y_4 :

$$y_4^{(c_1)} = y_2 + \frac{h}{3}\left[y_2' + 4y_3' + y_4'^{(c)}\right]$$

$$= 1.12 + \frac{0.1}{3}\left[0.6523 + 4(0.7979) + 0.9498\right]$$

$$= 1.12 + 0.1598 = 1.2798$$

Again $y_4'^{(c_1)} = f(x_4, y_4^{(c_1)}) = f(0.4, 1.2798) = 0.95$

Third approximation of y_4 :

$$y_4^{(c_2)} = y_2 + \frac{h}{3}\left[y_2' + 4y_3' + y_4'^{(c_1)}\right]$$

$$= 1.12 + \frac{0.1}{3}\left[0.6523 + 4(0.7979) + 0.95\right]$$

$$= 1.12 + 0.1598 = 1.2798$$

$\therefore$ The approximate value of $y(0.4) = 1.2798$

2. Using Milne's Predictor-Corrector method evaluate the integral of $5xy' + y^2 - 2 = 0$ at 4.4 given that $y(4) = 1$, $y(4.1) = 1.0049$, $y(4.2) = 1.0097$, $y(4.3) = 1.0143$

Sol. Here, $y' = \dfrac{2 - y^2}{5x}$, $h = 4.1 - 4 = 0.1$

x	y	$y' = f(x,y) = (2-y^2)/5x$
$x_0 = 4$	$y_0 = 1$	$y_0' = f(x_0, y_0) = 0.05$
$x_1 = 4.1$	$y_1 = 1.0049$	$y_1' = f(x_1, y_1) = 0.0483$
$x_2 = 4.2$	$y_2 = 1.0097$	$y_2' = f(x_2, y_2) = 0.0467$
$x_3 = 4.3$	$y_3 = 1.0143$	$y_3' = f(x_3, y_3) = 0.0452$
$x_4 = 4.4$	$y_4^{(P)} = 1.0187$	$y_4'^{(P)} = f(x_4, y_4^{(P)}) = 0.0437$
	$y_4^{(C)} = 1.0187$	

By Milne's predictor method,

$$y_4^{(P)} = y_0 + \frac{4h}{3}\left[2y_1' - y_2' + 2y_3'\right]$$

$$= 1 + \frac{4(0.1)}{3}\left[2(0.0483) - 0.0467 + 2(0.0452)\right]$$

$$= 1 + 0.0187 = 1.0187$$

$$y_4'^{(P)} = f(x_4, y_4^{(P)}) = f(4.4, 1.0187) = 0.0437$$

By Milne's Corrector method,

$$y_4^{(C)} = y_2 + h/3\left[y_2' + 4y_3' + y_4'^{(P)}\right]$$

$$= 1.0097 + \frac{0.1}{3}\left[0.0467 + 4(0.0452) + 0.043\right]$$

$$= 1.0097 + 0.009 = 1.0187$$

$\therefore$ The approximate value of $y(4.4) = 1.0187$

3. Given $y' = x(x^2+y^2)e^{-x}$; $y(0)=1$, find y at 0.1, 0.2 and 0.3 by Taylor's series method and compute $y(0.4)$ by Milne's method.

Soln Here, $y' = x(x^2+y^2)e^{-x}$; $y'(0) = 0$

$$y''(x) = e^{-x}(3x^2 + 2xyy' + y^2) - e^{-x}(x^3 + xy^2)$$

$$= e^{-x}(3x^2 + 2xyy' + y^2 - x^3 - xy^2)$$

$$y''(0) = 1$$

$$y'''(x) = e^{-x}(6x + 2yy' + 2xy'^2 + 2xyy'' + 2yy' - 3x^2 - y^2 - 2xyy')$$
$$- e^{-x}(3x^2 + 2xyy' + y^2 - x^3 - xy^2)$$

$$y'''(0) = -1 - 1$$

$$= -2$$

By Taylor's series method,

$$y(x) = y(x_0) + \frac{(x-x_0)}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{(x-x_0)^3}{3!} y'''(x_0) + \cdots$$

$$\therefore \quad y(x) = 1 + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(-2)$$

$$= 1 + \frac{x^2}{2} - \frac{x^3}{3}$$

$$\therefore \quad y(0.1) = 1 + \frac{(0.1)^2}{2} - \frac{(0.1)^3}{3} = 1.0047$$

$$y(0.2) = 1 + \frac{(0.2)^2}{2} - \frac{(0.2)^3}{3} = 1.0173$$

$$y(0.3) = 1 + \frac{(0.3)^2}{2} - \frac{(0.3)^3}{3} = 1.0360$$

x	y	$y' = f(x,y) = x(x^2+y^2)e^{-x}$
$x_0 = 0$	$y_0 = 1$	$y_0' = f(x_0, y_0) = 0$
$x_1 = 0.1$	$y_1 = 1.0047$	$y_1' = f(x_1, y_1) = 0.0922$
$x_2 = 0.2$	$y_2 = 1.0173$	$y_2' = f(x_2, y_2) = 0.1760$
$x_3 = 0.3$	$y_3 = 1.0360$	$y_3' = f(x_3, y_3) = 0.2585$
$x_4 = 0.4$	$y_4^{(P)} = 1.0701$	$y_4'^{(P)} = f(x_4, y_4^{(P)}) = 0.3499$
	$y_4^{(C)} = 1.0693$	$y_4'^{(C)} = f(x_4, y_4^{(C)}) = 0.3495$
	$y_4^{(C_1)} = 1.0693$	

By Milne's predictor method,

$$y_4^{(P)} = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3')$$

$$= 1 + \frac{4(0.1)}{3}\left[2(0.0922) - 0.1760 + 2(0.2585)\right]$$

$$= 1 + 0.0701 = 1.0701$$

$$y_4'^{(P)} = f(x_4, y_4^{(P)}) = f(0.4, 1.0701) = 0.3499$$

By Milne's corrector method,

$$y_4^{(c)} = y_2 + \frac{h}{3}\left[y_2' + 4y_3' + y_4'^{(p)}\right]$$

$$= 1.0173 + \frac{0.1}{3}\left[0.176 + 4(0.2585) + 0.3499\right]$$

$$= 1.0693$$

$$y_4'^{(c)} = f(x_4, y_4^{(c)}) = f(0.4, 1.0693) = 0.3495$$

Second approximation of y_4 :

$$y_4^{(c_1)} = y_2 + \frac{h}{3}\left[y_2' + 4y_3' + y_4'^{(c)}\right]$$

$$= 1.0173 + \frac{0.1}{3}\left[0.176 + 4(0.2585) + 0.3495\right]$$

$$= 1.0693$$

$\therefore$ The approximate value $\underline{\underline{y(0.4)}} = 1.0693$.